

The Invariance Principle

Thomas Kidane

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1 Introduction

This is a collection of attempts to solve the problems in A.Engels Problem Solving Strategies book, chapter 1. Unfortunately I don't have the notes of me solving problems upto 19, and I started keeping in non-latex starting from problem 20, and I will solve them in the computer starting from problem 24.

2 non-Latex(and source for other solutions)

Techniques for Solving

2.1 Making Invariants

- Get the coefficients for making an invariant by checking the recursive steps with zero and one and checking how the coefficient should behave.

2.2 Problem 20

A game for computing $\gcd(a, b)$ and $\text{lcm}(a, b)$.

We start with $x = a$, $y = b$, $u = a$, $v = b$ and move as follows:

- If $x < y$, then set $y \leftarrow y - x$ and $v \leftarrow v + u$.
- If $x > y$, then set $x \leftarrow x - y$ and $u \leftarrow u + v$.

The game ends with $x = y = \gcd(a, b)$ and $\frac{u+v}{2} = \text{lcm}(a, b)$. Show this.

- This feels like the Euclidean algorithm.
- Let $x = id$, and $u = id$ where $d = \gcd(a, b)$.
- Let $y = jd$, and $v = id$ where $d = \gcd(a, b)$.
- $\gcd(i, j) = 1$.
- The algorithm goes:

– If $i < j \rightarrow j = j - i$, and $v = \dots$

- Look for an invariant: Since $\text{lcm}(a, b) / \text{gcd}(a, b) = ab$, see if an invariant can be generated using this fact.
- What happens to the difference between i, j and if they are coprime?
- The invariant found in the book is $xv + uy = ab$.
- Multiplication techniques where terms getting multiplied are not necessarily ‘close together’ might be useful.

2.3 Problem 21

Three integers a, b, c are written on a blackboard. Then one of the integers is erased and replaced by the sum of the other two diminished by 1. This operation is repeated many times with the final result 17, 1967, 1983. Could the initial numbers be:

1. $(2, 2, 2)$
2. $(3, 3, 3)$

Transformation: $(a, b, c) \rightarrow (b + c - 1, b, c)$

- The transformation preserves the two other numbers.
- Attempting to find an invariant in this situation.
- Observation: Doing the operation twice to the same spot does nothing.
- Chain of operations always involves choosing different spots each time.
- The last chosen number becomes the biggest.
- The difference between $\max(a, b, c)$ and the two lowest ones is 1.
- To reach $(17, 1967, 1983)$, reduce it to smaller steps recursively.

2.4 Problem 22

There is a chip on each dot in Fig. 1.6. In one move, you may simultaneously move any two chips by one place in opposite directions. The goal is to get all chips into one dot. When can this goal be reached?

- If n is odd, the middle point can be used to aggregate all chips.
- If n is even, the situation is different.
- Assume a point as a middle where all chips can aggregate.
- The invariant could be the distance to this point under modulo 2.
- The sum distance from the supposed middle point to all other points remains invariant under the allowed moves.
- The proof shows that the invariant isn't 0 at the start, thus proving the impossibility.

2.5 Problem 23

Start with n pairwise different integers $x_1, x_2, \dots, x_n$, ($n > 2$) and repeat the following step:

$$T : (x_1, \dots, x_n) \rightarrow \left(\frac{x_1 + x_2}{2}, \frac{x_2 + x_3}{2}, \dots, \frac{x_n + x_1}{2} \right).$$

- The task is to show that $T, T^2, \dots$ finally leads to non-integral components.
- The operation is essentially averaging in a circular manner.
- The initial step already produces non-integer values since all integers are pairwise different.
- A search for an invariant under this transformation might reveal why non-integral components must eventually arise.
- I tried to make a linear transformation A, whcih encapsulates this process.
- and I tried to show that applying this probesss doesn't get you towards the constant
- vector, which is just a vector with entries begin the mean if of the values.
- I got the solution to work if n is odd, then the resulting matrix is non-singular?
- Sorry, the matrix doesn't have an eigenvalue of 1, when it is even, and instead has null space when it is even.
- But instead the book, first assumes that it becomes integral and shows that in a finite
- number of steps, the vecotr becomes all a's.
- Pf: After it has become integral then the distance between the maximum and the minimum decreases,
- in fact if the minimum appears m, times, after m iterations it decreases, and vice versa for minim.
- Then it goes to show that you get a,a,a, and it goes to show that the previous iteration is all the
- same in case of odd, because the odd numbers are equal and the even equal equal.
- You can also show this is the case in even, since the sum of all evens are the same as the same of all odds(each number is the average of the two numbers next to it)
- So either way you can't have equality, if you didn't start out with equal numbers
- Second proof uses permutation inequality, should probably check that later, but it will probably come up in the book anyway.

2.6 Problem 24

Start with an $m \times n$ table of integers. In one step, you may change the sign of all numbers in any row or column. Show that you can achieve a nonnegative sum of any row or column. (Construct an integral function which increases at each step, but is bounded above. Then it must become constant at some step, reaching its maximum.)

Solution

Weird for the book to give me hints, but okay. Isn't this question also in olympiad combinatoris? Just wait for the day I become furnished in the art of problem solving. Ok let's go back to solving the problems.

Is the question asking, if I can achieve it for only one at a time? obviously that is true, if it is only for one, I guess it is asking me for all at the same time. Can I make a process that increases monotonically like the problem said? If I only do a greedy algorithm where I flip the the most negative row or column, will it effect the entire sum more than the change in that row/column will? Basically I am trying to use the sum of the rows and columns as my function and increase monitncally?

$$\sum \text{rows} = \sum \text{columns}$$

Asssume the smallest negative row is r_1 which smaller than all individual rows or columns. The sum of columns is sum all the rows except v_1 and v_1 . If I change the sign of v_1 , then not only will the sum of the rows increase the sum of all the columns will increase. So the function which is the sum of all the entries in the table is the function we are looking. This function is bounded above by the sum of the absolute value of all the numbers and therefore the algorith which picks the row or column with negative entries and flips them is terminating.

2.7 Problem 25

Assume a convex $2m$ -gon $A_1, \dots, A_{2m}$. In its interior we choose a point P , which does not lie on any diagonal. Show that P lies inside an even number of triangles with vertices among $A_1, \dots, A_{2m}$.

Solution

Let me try on a sqaure.

$$\begin{array}{c} A_1 \dots A_2 \\ \dots\dots\dots \\ \dots P \dots\dots\dots \\ A_3 \dots A_4 \end{array}$$

In the picture, P doesn't lie in between the diagonal A_1 and A_4 . Yes it is infact in 2 triangles, specifically the traingles that don't include A_4 and A_2 .

Could it be that P , is in precisely half the number of triangles. Since the question said that this property works for any P that isn't on a diangonal, that means it works on all points, how could I use that?

What if I chose a point that is very close to one of the vertices. Then all diagonals "pass" over it. So basically all the triangles that have, actually name the vertex it is close to v_1 .

So all triangles that have v_1 a corner point satisfy. Since any diagonal I choose between the other points will make a triangle.

The number of triangles will become $\binom{2m-1}{2} = \frac{(2m-1)!}{2!(2(m-1)-1)!} = \frac{(2m-1)*(2m-2)}{2} = (m-1)(m-1)$. Is this even? if m -even then you get odd? Oh, is not necessarily true that the point will be in all the triangles that have v_1 as a vertex.

Could show that each diagonal hits every point twice? Let's get back, first are there even number of triangles?

$$\binom{2m}{3} = \frac{2m!}{3!(2m-3)!} = \frac{2m * (2m-1) * (2m-2)}{6}$$

Since there is a 4 in the numerator, it is guaranteed that the number of triangles is even. Will there be regions covered odd number of times. Actually the number of triangles is divisible by 4, because you can take 2 from both $2(m)$ and $2m-2$.

Now consider what happens when you have a triangle. You have some points outside of it, and the inverse of this triangle creates?

There are three kinds of triangles depending

Ok I looked at the proof, and I should rework my idea on the moving the point that is very close to the point being very close to an edge. Then the point is in $2m-2$ triangles, when the point moves from one diagonal to another, it suddenly stops being in the diagonals on side to all the triangles on the other side of this diagonal. the change in number of triangles is $a-b = a+b = 2m-2 \pmod{2}$, where a is the number of triangles having a side that is this diagonal and vice versa for b .

2.8 Problem 26

Three automata I, H, T print pairs of positive integers on tickets. For input (a, b) , I and H give $(a+1, b+1)$ and $(a/2, b/2)$, respectively. H accepts only even a, b . T needs two pairs (a, b) and (b, c) as input and yields output (a, c) . Starting with $(5, 19)$ can you reach the ticket (a) $(1, 50)$ (b) $(1, 100)$? Initially, we have $a, b, a < b$. For what n is $(1, n)$ reachable?

Solution

Ok, so I adds one to both numbers and H divides them by 2. T takes two pairs, and gives you the number that is not found twice in both pairs. Does the question imply that you only have one pair and can't use the same pair twice without making another one? if I use one pair? do I lose it? that would imply that I only have one ticket no matter what and can't ever use machine T. So I get to keep my original pairs,

Let me take a look at the ending condition, since it is $1, 50$. I have to have used a divide on my last operation so it becomes $2, 200$. I also see that if the difference between a and b is d , initially then the division operation can only be done as many times you can divide d by 2, since each time it divides it by two and the difference has to be even for the two numbers to be both even at the same time. So the number of times I can use machine H, is limited to once. So the can I get to $2, 200$. Unfortunately no. So it is impossible with machine I and H, alone.

Will T change this problem? If I want to use it then b has to be the same to a . For the two pairs I will make in this process. But does T even change the difference between

a and c? the difference between a and b is off the form $odd * 2^n$, then the new difference will also have this odd term. You can't make this odd term lower. than the original odd, part, but I guess you can make it bigger. What happens if you make it bigger. The new difference is still divisible by the odd part. In 1 and 50 the difference is 49, which could be possible since it is divisible by 7, but definitely not for 1 and 100, since the difference isn't divisible by 7.

But does divisibility imply that I can get to this condition. Since $a < b$ the operations maintain the order of the elements. I. Actually I think I should take a look at the problem more, because I don't think I know the rules properly enough. Nothing new. Let me try to get to 1,50

5, 19 $\rightarrow$ 3, 10 first add one and then divide

3, 10 $\rightarrow$ 19, 26

5, 19 19, 26 $\rightarrow$ 5, 26

5, 19 $\rightarrow$ 26, 40

5, 26 26, 40 $\rightarrow$ 5, 40 5, 19 $\rightarrow$ 40, 54

5, 40 40, 54 $\rightarrow$ 5, 54

Ok the difference is 49 but I can't divide it. Actually I can use the same process right here to increase the difference, while keeping 5, constant. Then make it so that the difference is $49 * 2^3$. Then increase them both by 3, and keep dividing away and then at the end you get 1 and 50.

Ok in a weird twist of turns a and b are both solved. I can essentially use this technique to make 1 and n as long as $n = 7 * 2^x$ where x is any non-negative integer.

Apparently, the old page doesn't post solutions after this problem. So I guess I am cooked if I don't solve these problems by myself. I am tired I am going to look at the problem and sleep, my goal is 10 problems a day.

2.9 Problem 27

Three automata I, R, S print pairs of positive integers on tickets. For entry (x, y) , the automata I, R, S give tickets $(x - y, y)$, $(x + y, y)$, (y, x) , respectively, as outputs. Initially, we have the ticket $(1, 2)$. With these automata, can I get the tickets:

1. $(19, 79)$
2. $(819, 357)$

Find an invariant. What pairs (p, q) can I get starting with (a, b) ? Via which pair should I best go?

Solution First invariant I see is the gcd, since $\gcd(x, y) = \gcd(x, y - x)$. The same is unfortunately not true for the sum. S swaps the numbers. X is invariant under gcd of the numbers. I am going to be bed. See you tomorrow.

As I was walking today, realized that this is just euclidean algorithm. You can start from any pair (a, b) and keep doing the euclidean algorithm to go down, to $(1, d + 1)$

2.10 Problem 28

n numbers are written on a blackboard. In one step you may erase any two of the numbers, say a and b , and write, instead $(a + b)/4$. Repeating this step $n - 1$ times, there

is one number left. Prove that, initially, if there were n ones on the board, at the end, a number, which is not less than $1/n$ will remain.

Solution

Now what should the invariant be? How do I decrease the numbers fast enough? The best choice looks like to just choose two, get one over two, then choose two one over two to make $1/4$. This process would respect the bounds set by the problem.

Can I prove that I can't decrease it faster than this? Wait if I start with 2 ones, I can't smaller than $1/2$. If I start off with 3 ones. I can't get any lower than $1/2$.

Let me make a function. The smallest number $f(x)$, the smallest number I can achieve with x numbers.

$f(0) = 0, f(1) = 1, f(2) = 1/2, f(3) = (f(2) + f(1))/2 \dots$ Everytime I combine numbers two numbers, $f(x)$ and $f(y)$, I am trying to make $f(x + y)$. I want to prove that this is a lower bound. Assume you try to make $f(n)$, then in the last operation you combine $f(n - x)$ and $f(x)$.

If I prove the lower bound on this function, I am done. So the function has the recurrence.

$$f(n) = \min_{1 \leq i \leq n-1} \frac{f(i) + f(n-i)}{2}$$

When does $f(n)$, reach its minimum, precisely when $f(i) + f(n-i)$ reaches its minimum.

Now we have the bound for $f(1)$ and $f(2)$. And these satisfy the hypothesis. Assume that the hypothesis is satisfied for some n .

$$f(n) = \min((\frac{1}{x} + \frac{1}{n-x})/2) = ((n-x) + x)/2 * (x * (n-x))$$

$$f(n) = \frac{n}{2 * (nx - x^2)}$$

Since the numerator is fixed, we just have to maximize the denominator, and the max of $nx - x^2$, is reached at $2ax - b$ from calculus, which for this case becomes $a = -1, b = n, -2x - n = 0 \rightarrow x = -n/2$ I guess didn't restrict it to be positive.

I just checked wolfram and the maximum does indeed become $x=n/2$, What happened? Oh my formula is wrong it $2ax + b = 0 \rightarrow -2x + n = 0 \rightarrow x = n/2$.

Ok now I got what I wanted, let's show that this is smaller than the reciprocal.

$$f(n) = n/(2 * (n^2/2 - n^2/4)) = \frac{n}{2 * (n^2/4)} = \frac{n}{n^2/2} = 2/n$$

So the minimum is $2/n$. Is this even true? Doesn't seem so for $n=2$. What went wrong?

Oh the original operation was to divide by 4, not 2. How does that affect our equation? for $f(2) = (1 + 1)/4 = 1/2$. This doesn't affect the original equation, it just makes it

$$f(n) = \min((\frac{1}{x} + \frac{1}{n-x})/4) = ((n-x) + x)/4 * (x * (n-x))$$

So everything is divided by 2 and $2/n$, becomes $1/n$. Finally done.

After looking at the book, it had a very clever invariant that involves the inverses of the two numbers being added. $1/a + 1/b \geq 4/(a+b)$, which comes from

$$\begin{aligned}
(a-b)^2 &\geq 0 \\
(a+b)^2 - 4ab &\geq 0 \\
(a+b)^2 &\geq 4ab \\
(a+b)/2 &\geq 2ab/(a+b) \\
(a+b)/(ab) &\geq 4/(a+b) \\
1/a + 1/b &\geq 4/(a+b)
\end{aligned}$$

The fact there was 4, or 2 should have hinted at you to try inverses as an invariant. Then this means that the sum of the reciprocals is strictly decreasing. Since the original reciprocals is n , then $1/S \leq x$

2.11 Problem 29

The following operation is performed with a nonconvex non-self-intersecting polygon P . Let A, B be two non-neighboring vertices. Suppose P lies on the same side of AB . Reflect one part of the polygon connecting A with B at the midpoint O of AB . Prove that the polygon becomes convex after finitely many such reflections.

Solution

What the hell is the question asking? SO I have a nonconvex non-self intersecting polygon called P . A and B are two non-neighbouring vertices. What does it mean for P to lie in the same side of AB ?

There is the line AB ? Ok I think what it is doing is turning the nonconvex part into convex.

Ok the invariant here should be the number of interior angles greater than 180.

I have a few questions I need to prove, to make this work

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The number of points with interior angle greater than 180 degree, decrease This procedure doesn't create any more This procedure only affects this processes.

Ok first, thing is that you can only execute this process, if all the vertices between A and B have interior angles greater than 180 degrees. Or atleast one of them has to.

How do I prove this? Since A and B , are non-neighboring, then there are vertices in the middle of them Since P lies on one side of this, then there are no vertices on the other side. You have two paths between A and B , since it is a closed shape. Since the polygon is non-intersecting this paths are distinct, and therefore one path is always closer to the side AB , than the other. Take the one that is closer, there must exist a point that has an interior angle greater than 180, because the path has to walk a straight line but it takes a turn towards one side, so it has to turn back and when it does, it creates an interior angle greater than 180. After the operation, this 180degrees is cut. No other angle is changed except the ones on both sides of the paths, and they both become less than 180 after this operation. So the number of 180 degree can only decrease, and you can

only do this operation if there are paths where atleast one vertex has an interior angle of more than 180 degrees. So the number of operations is finite and at the end you will get a polygon whcih is non-self intersecting and has no internal angeles greater than 180 degrees.

After taking a look at the book, it said the area is stricly increasing and is therefore finite.

2.12 Problem 30

Solve the equation $(x^2 - 3x + 3)^2 - 3(x^2 - 3x + 3) + 3 = x$

Solution

So first, thing I tried is $x = 3$, which gets me $9 - 9 + 3 = x \rightarrow x = 3$. Which is infact correct. So I guess I need to find the other. Now I have a cubic equation So I am guarenteed another real root, and once I get that I can find the other two roots using the quadratic equation.

So how can I find the other real root? Lowkey the equation looks like a repeat of itself. For example if you take the quadratic to be x , you get $x^2 - 3x + 3 = 0$. Will $x = 0$, work? $9 - 9 + 3 = 0$, which is wrong. Can I factor out?

Does it look factorizable? I am going to try to make it easier to type this out.

$$\begin{aligned}y &= x^2 - 3x + 3 \\y^2 - 3y + 3 &= x \\y^2 - 3y &= x - 3 \\y(y - 3) &= x - 3\end{aligned}$$

Can I try to divide y by x . OK I got $q(x) = x + \frac{3}{x-3}$. Isn't there theorem that the remainder becomes 0, when you divide by the roots or something. I don't know if that will work in this situation.

Can I try to come to idea that it is copy of itself? How can I use that? Why did that pop out to me? Can I first try to factor out y ? Ok, that has the complex square roots, $x = \frac{3 \pm i\sqrt{3}}{2}$. I don't think that will help.

Ok I got a root when it was $x = 3$, and that coincided with with roots of the equation, one the right. Now I need to find points where they become equal again. The one on the right is just a linear function. So if I know what x , then it is easy to figure out. The quadratic is also centered around $3/2$, maybe I can try $3/2$? So let me pull up desmos to graph y . $y(3/2) = 3/4 \rightarrow y(3/2) - 3 = \frac{-9}{4}$.

$$y(y - 3) = 3/4 * -9/4 = -27/16$$

, and $x - 3 = -3/2$. Can I try to do derivatives?

Wait this unit is about invariants, so I should look for invariants. Do I see any invariants? What about $x = 1, \rightarrow 1 - 3 + 3 = 1$, Awesome I have 2 of the roots. Shoud I brute force it right now?

Maybe I can extract some other values throuh observation? First the final equation needes to be able to be decompsted into this form. What if I look at the constant term? Could one of them be a repeated root? The constant term is a multiple of all the roots, and it is 3, in this case. Let my check if $(x - 1)^3(x - 3)$, could be the equation.

Lowkey I am just gonna brute force it. I was right it was just a repeated root with $(x-1)^3(x-3)$

Ok after reading the solution, it was just that $f(f(x))$, I guess whne it popped out to me that the similarity, was the it was just the function plugged into itself. So it would be just the points that are invariant under the function itself. Nice trick!

2.13 Problem 31

Let $a_1, a_2, \dots, a_n$ be a permutation of $1, 2, \dots, n$. If n is odd, then the product $P = (a_1 - 1)(a_2 - 2) \dots (a_n - n)$ is even. Prove this.

Solution Should I expand out the terms? I feel that it is simpler to prove that, one of the terms has to be even? because if even one of the number is even then the entire expression is even.

Assume the opposite, so all the expressions have to be odd. For the difference to be odd, the numbers have to be odd and even. There are $\frac{n+1}{2}$, odd numbers and $\frac{n-1}{2}$, even numbers. If you pair up each odd number with each even number, and vice versa for the values from the other permutation then you are left, with two odd numbers paired up that makes it even.

If n was even you could have done this pairing just fine, and you would be done. I feel that there is something about invariance here, the pairings always have the form of one odd and one even, is that the invariance? I guess that is it.

After reading the book, it just takes the sum of the factors, and since the sum must be equal to 0, this contradicts the fact that there are odd number of odd numbers. I guess it is more elegant than my pairing up approach.

I think for the future, I should look for ways to reduce the problem, or look at other cumulative properties, so that I can make my proofs based on that, like sums and properties that utilize all elements involved in the problem.

2.14 Problem 32

Many handshakes are exchanged at a big international congress. We call a person an *odd person* if he has exchanged an odd number of handshakes. Otherwise he will be called an *even person*. Show that, at any moment, there is an even number of *odd persons*.

Solution

Time to look at cumulative properties!

Each handshake is counted twice, one for each party involved. So the total number of handshakes at any one point is even. Assume there are x odd persons and y even persons.

Actually better, to label x as the total number of handshakes all odd persons have made. Also label y , as the total number of handshakes made by all even persons. Since $x + y$, is the total number of handshakes made, this number has to be even. Since y is even then x , to be even.

Since the sum off odd number of odd numbers is odd, you can't have odd number of odd people. Done!

The books are argument is that, the number of odd persons, can't change parity in all cases of handshakes, and since it started at 0, it must also be even. Nice.

2.15 Problem 33

Start with two points on a line labeled 0, 1 in that order. In one move you may add or delete two neighboring points (0, 0) or (1, 1). Your goal is to reach a single pair of points labeled (1, 0) in that order. Can you reach this goal?

Solution

Ok, after looking at the question after so long, I understand what it is saying. You have a list, that started as 0,1. You can add or remove two neighbouring values 0,0 or 1,1. Can I get to 1,0.

When you add and remove, aren't you doing the same thing? Maybe I can make an argument on the number of inversions? like the number of inversions is 0, when you started and you can't change the parity of the inversions? Actually let's see if the parity of the inversions is conserved when I add and remove elements.

At the start the parity of the number of inversions is 0. Let's you add one of the two pairs at any arbitrary position. If it is 1,1, then look at all the 0, next to this pair. The parity increases by twice this number since it increases the inversion count for both 1's. Likewise, for the 0's, look at the number of 1, to the left and the parity increases by twice this amount.

So the parity is conserved for adding and with the same argument for removing as well. I guess I solved it with CS knowledge. The book does the same thing.

2.16 Problem 34

Is it possible to transform $f(x) = x^2 + 4x + 3$ into $g(x) = x^2 + 10x + 9$ by a sequence of transformations of the form $f(x) \rightarrow x^2 f\left(\frac{1}{x} + 1\right)$ or $f(x) \rightarrow (x - 1)^2 f\left(\frac{1}{x-1}\right)$?

Solution

What invariants come to my mind? Can I look at the constant terms? What about the the power of the highest term? Wait is that even conserved?

Ok the coefficient of the x^2 , is always increasing. Could that be my invariant. Should I look at points that are invariant under this transformation.

Our function can be decomposed into $f(x) = (x+1)(x+3)$, $g(x) = (x+1)(x+9)$. They don't share a root. Could I show that this root is conserved under both transformations? or the inverse is conserved.

If I plug -1, into the first function I get $f(-x + 1) = f(0)$. So the -1, gets mapped to 0? What about -3? $9f(x/-3 + 1)$. What about the other transformations? When I plug -1, $4f(1/-2)$, which is some number.

Since at the end, the transformation always give me a quadratic, I should try to show that the quadratics, they give have properties, that I can't have. Will it ever map the roots of f to the roots of g . At the end, the roots should be -9, and -1.

At the end the polynomial must have the form of $x^2 l(x)$, or $(x - 1)^2 l(x)$. Where $l(x)$, is the result of applying the previous transformations. But I can plug in 1, or 0 in both cases to get 0!!!

Ok, so matter what happens the middle the functions have to be of this form, I can always plug in 0, 1 to get a root. But These aren't root of the polynomial equation I want to get so I can't ever get $g(x)$.

Ok, the book said the transformation preserved the discriminant, and said the two polynomials have different discriminants. I guess my argument not the intended?

2.17 Problem 35

Does the sequence of squares contain an infinite arithmetic subsequence?

Solution

Two simple solutions. Assume there was an infinite sequence between the square numbers. Their difference in between successive square numbers is a strictly increasing sequence, so for any finite difference of the arithmetic sequence, there will be a point after which there are no more terms in that sequence since the difference between successive elements will be greater than the difference.

The explanation from the book is very nice and shows the technique of proof of contradiction by showing an infinitely descending chain of solutions.

$$\begin{aligned}(a_2^2 - a_1^2) &= (a_3^2 - a_2^2) \\ (a_2 - a_1) * (a_2 + a_1) &= (a_3 - a_2) * (a_3 + a_2) \\ (a_3 + a_2) > (a_2 + a_1) &\rightarrow (a_2 - a_1) > (a_3 - a_2) \\ (a_2 - a_1) > (a_3 - a_2) &> (a_4 - a_3) > (a_5 - a_4) \dots\end{aligned}$$

You can't have an infinite descending sequence of positive integers.

2.18 Problem 36

The integers $1, \dots, n$ are arranged in any order. In one step you may switch any two neighboring integers. Prove that you can never reach the initial order after an odd number of steps.

Solution

We just need to prove it for the identity permutation case, because we can relabel any sequence like the identity.

So after seeing I immediately think of counting the number of inversions. Hopefully by neighboring it doesn't also mean that the last and first element are also neighbors, since my proof to follow has requires this assumption.

Ok, a swap between adjacent elements can only remove or add one inversion to the permutation, if you include the operations required to first increase and decrease the number of inversions that will always be even. This is because any move in the starting state will increase the number of inversions by one, and in the final state that inversion has to be gone, and it requires one move to put fix it, and a move can only fix one inversion, so you create an inversion and for every inversion that is created you have to do one move to fix that inversion, so in total the number of moves is always even.

The book's argument is that any move always changes the parity of the number of inversions, which is a more clearer argument.

2.19 Problem 37

One step in the preceding problem consists of an interchange of any two integers. Prove that the assertion is still true.

Solution

Well there goes my proof from the last problem. Wait it said only one step is just an interchange of any two numbers. I need to prove that this move still can't keep the parity of the inversions the same.

Ok, what are the possible cases that can happen. If the two elements are next to each other then there is nothing to prove, so assume there is a gap between them.

Let's see what happens to the inversions for each individual element in the range. The two numbers being swapped can both be bigger than this person, or smaller, or both. If both numbers are bigger, the parity doesn't change. If both are smaller, then the parity doesn't change. If one is bigger and the other is smaller, either the parity increases by 2 or decreases by 2. So the parity doesn't change.

What about the actual elements being swapped? They do indeed increase or decrease the parity by 1. So the parity always changes parity in each move, so you always need an even amount of moves to return to the original form.

2.20 Problem 38

The integers $1, \dots, n$ are arranged in order. In one step you may take any four integers and interchange the first with the fourth and the second with the third. Prove that, if $\frac{n(n-1)}{2}$ is even, then by means of such steps you may reach the arrangement $n, n-1, \dots, 1$. But if $\frac{n(n-1)}{2}$ is odd, you cannot reach this arrangement.

Solution

Lowkey the expression looks so similar to the total sum, but it should be one higher to be the total sum of elements. Oh, this is actually the number of inversions in the result. Since the last element should come before all the elements after it, it contributes $n-1$ inversions. And so on with the other elements leading to the formula.

I guess I just need to prove that the move preserves parity. Since from the last question, I know that every move changes parity, if you do 2 you get the same parity you started out. If the total number of inversions is odd then you can never get to a parity of 0.

2.21 Problem 39

Consider all lattice squares (x, y) with x, y nonnegative integers. Assign to each its lower left corner as a label. We shade the squares $(0, 0), (1, 0), (0, 1), (2, 0), (1, 1), (0, 2)$.

- (a) There is a chip on each of the six squares.
- (b) There is only one chip on $(0, 0)$.

Step: If (x, y) is occupied, but $(x+1, y)$ and $(x, y+1)$ are free, you may remove the chip from (x, y) and place a chip on each of $(x+1, y)$ and $(x, y+1)$. The goal is to remove the chips from the shaded squares. Is this possible in the cases (a) or (b)? (Kontsevich, TT 1981.)

Solution

I spend these last two days trying to understand what the question was even saying. But I guess I understand now that it is saying.

So it is basically saying what it is saying. So it has the lattice points of half-square spanning from $(0,0)$ to $(2,2)$. And the moves I can do are stated above. What invariant do I see?

Let me first try to do it. So I move the chip at $(0,2)$, do put chips at $(0,3)$, and $(1,2)$. Then I will try to move the next one in the middle but I will have to move this two to move that one.

The number of chips on the plane is always increasing. So each time I make a move I am basically giving the square an initial velocity and of 1 up and 1 right and take snapshots and merge them.

So what other invariants do I seem assume that a chip is placed at (x,y) , I want to prove that there will either be a point here or $(x+1,y+1)$. Is that even possible? Let me consider another fact?

Each square in the column y will have one square on it. Because each time you do a move, the point will replicate to the right not to the up and again. I think I could prove this inductively

The point (x,y) , will not be occupied again since all the other replications will be one up or right and they cannot go back, now assume a point is reached twice. Oh this is simply not true.

Ok now what do I do? It feels like no matter what I do the points keep hitting each other. And I cannot move this point without moving this point but I cannot do that.

Let me make an invariant that is defined on this space. Maybe specifically on this half-square. What is conserved or monoinvariant?

Ideally whatever invariant I have it must treat $(0,1)$ and $(1,0)$ the same, and $(2,0)$ and $(0,2)$ the same. What would happen $(1,1)$? Should I treat it as?

Well the difference between the coordinates is conserved? Or atleast you get twice the difference? Isn't it weird that all the points either move to the right or up?

Could I make an invariant on that? What is the main difference between the squares not shaded state and the squares shaded part?

What about the difference between the highest y and most right x ? Not an invariant

My invariant would have to satisfy? What happens if two points are on (x,y) and $(x+1,y-1)$. I would like these 2 points to somehow result in the previous point?

What if I go back? Assume you do a move and that you get a free square. This move could have happened on the diagonal and WLOG on the top square. If you did it on the top square then that means there is no points above it? Because there can't be anymore on this column you didn't make any moves else. And it has to be that $(1,2)$ had to be free. And that would imply that there is a point above, because in the second step, there is always a point above. But a above also implies a point on the same row as well. Somewhere there had to be a point and that surpassed up, $(1,2)$. OK I looked at the book solution and it had a very creative invariant.

Assign each chip the weight $\frac{1}{2^{x+y}}$. Then each move will conserve the weight of the chips on the board. The first one is impossible because the weight of the shaded region is $2+3/4$, but the weight of the result of the board is only $1+3/4$. So we get a contradiction. As for the other one, since the weight of the lone piece is 1, and the weight that is on the rest of column $x=0$ is only $1/8$ and likewise for the row $y=0$. The weight of the rest of the table is $3/4$, but you can't cover this much space in finitely many moves. So another contradiction.

What could I have done better.

- I shouldn't have stayed away from algebraic arguments, and you can be as creative as you want with your invariants, it doesn't necessarily have to be the case that you invariants are on the objects themselves.

2.22 Problem 40

In any way you please, fill up the lattice points below or on the x -axis by chips. By solitaire jumps try to get one chip to $(0, 5)$ with all other chips cleared off. (J. H. Conway.) The preceding problem of Kontsevich might have been suggested by this problem.

A solitaire jump is a horizontal or vertical jump of any chip over its neighbor to a free point with the chip jumped over removed. For instance, with (x, y) and $(x, y + 1)$ occupied and $(x, y + 2)$ free, a jump consists in removing the two chips on (x, y) and $(x, y + 1)$ and placing a chip onto $(x, y + 2)$.

Solution

I don't think this is true. It simply takes too many points. Even I have infinite number of points I couldn't I jump to this point.

Let me look for invariants in this problem. What could it be? I just need to get one chip to $(0,5)$, because once I get to there I can get rid of all the chips in reverse that I didn't need and I am only left with that chip.

So first assume that this chip does indeed reach that position. Then WLOG generality its last step must have a am up jump. So it must have jumped over a piece one below at $(0,4)$. First of all I can get to $(0,1)$. I can infact fill all of it up. but I can an infinite number of them. But I won't have anything below it. After I have this entire row at $(0,1)$, then what do I do? I can't move, I need free space.

My invariant requires free space? Can I get to $(0,2)$. I need a kind of conveyor belt. Where I can points that only go up, and I have points on the side where they can fill in the gap.

To get to $(0,5)$, I have to jump over $(0,4)$, from $(0,3)$. I can comfortably jump to $(0,2)$. I just have one jump to $(0,1)$, it side fill the gap in. Then it jumps to $(0,2)$. At this point the first point below $(0,2)$ has height -2. Is this like the previous question but now in 3s can twos?

Like a chip on the $y=1$ equal to chip on line $y=-1$ and $y=0$ summed. What about going sideways? Actually just decreases the points in the system. Now what would be the recurrence relation? I definetly want a point on a higher position to have a higher value than one on a lower position.

$$f(y) = f(y - 1) + f(y - 2)$$

This looks like the fibonacci sequence. I also have to penalize for going sideways to, so the recurrence would be a term that includes how far to the side it is as well. But I primarily want to punish, go sideways and going down, which is just nonsensical, or maybe doesn't matter.

Let's say the wait of a chip is worth some amount.

$$x + x^2 = 1$$

$$x = \frac{-1 \pm \sqrt{5}}{2}$$

We will take the positive root ϕ . This value is indeed less than one. So if I take the infinite geometric series it should converge. This value is indeed less than one. So if I take the infinite geometric series it should converge.

$$w(x, y) = \frac{1}{\phi^{-y+\|x\|}}$$

So the point (0,5) has weight ϕ^{-5} . How much does the column $x = 0$ have? $1 + \frac{1}{\phi} + \frac{1}{\phi^2} + \dots = \frac{1}{1-\phi}$. On either side I have

$$\begin{aligned} l &= \frac{1}{1-\phi} \\ S &= l + 2 * \phi^{-1} * l + 2 * \phi^{-2} * l + \dots \\ S - l &= 2 * (l - 1) * \phi * l \\ S &= 2 * (l - 1) * \phi * l + l \end{aligned}$$

Now I want to compare this to S to ϕ^{-5} .

$$S - \phi^{-5} = 2 * (l - 1) * \phi * l + l - \phi^{-5}$$

I would assume this is negative or something. Making it impossible.

After reading the book I found that my weight function was wrong. It should be $w(x, y) = \phi^{-y+\|x\|}$ since I want the weight to go down. A mistake on my part, the book actually used $w(x, y) = \phi^{\|x\|+\|y-5\|}$ as its weight function to normalize the value. Which I should do next time. It was kind of hard to estimate this quantity without using a calculator. But it was a nice idea to put the invariance as a function that is similar to the action. I guess this is a game of finding functions that are invariant, trying to make them myself. The book also used a finiteness argument at the end. It is okay if my invariant function only meets the final condition equally, if I can use the finiteness condition to go that last part. So disprove prove that it has to be infinite to be valid and use the finiteness property to invalidate it.

2.23 Problem 40

We may extend a set S of space points by reflecting any point X of S at any space point A , $A \neq X$. Initially, S consists of the 7 vertices of a cube. Can you ever get the eight vertex of the cube into S ?

Solution

What is the question asking? Ok, thanks to chatGPT I now know what reflection off a point means.

To finally reflect of a point onto the last point, means that point was the reflection of another point, obviously. Let's try to derive the formula for reflection. I want to; reflect (x, y, z) across (a_1, a_2, a_3) . The new point will be $(a_1, a_2, a_3) - (- (a_1, a_2, a_3) + (x, y, z)) = 2(a_1, a_2, a_3) - (x, y, z)$, can I get an invariant out of this?

My function should be consistent with this invariant. Ok to make it more concrete, assume WLOG that the missing point is (1, 1, 1). Let's make an invariant.

If I had an invariant I would want it not to change for the new point which will be the result of the reflection. Actually let the point be that is missing be (0,0,0), since that will be easier as an ending state. The function I want should be invariant under the above mentioned transformation. What if it is just euclidean distance? Wait what about L_1 norm?

If the function spits out the same number for each point the above transformation looks like $2x - x = x$? What function is conserved when do I this operation? Actually it also has to give these same valus for the other 7 points. What are the other 7 points?

(0, 0, 1)
 (0, 1, 0)
 (0, 1, 1)
 (1, 0, 0)
 (1, 0, 1)
 (1, 1, 0)
 (1, 1, 1)

These look like bits don't they? Will it help if I look at them in base 2? Only the first row is odd. So in base 2, the second and third columns conserve their parity? Actually is the parity conserved in all of them? Nop. Should I make function on this? Better focus my efforts elsewhere. What if I gave each point a nonzero value and then make it like a gcd problem where you can't get to 0? What if I try to get to (0,0,0)? I have the formula above? With these values and the operation defined above(Reflection) can I get to (0,0,0). What about a monovariant? Now that I think about it all the points will integral points, so I could use Number Theory on it.

Could I describe it as a diophantine equation? What if I multiply each value by a prime number, like 2,3,5. So the value of a point is $a_1 * 2 + a_2 * 3 + a_3 * 5$, now what does it mean for two values to be equal? Or what does it mean to get 0? Can I even get 0? In this? Well yes, 2 can be described as a linear combination of 3 and 5. What about if I make.

Actually now that I think about I can exploit this divisibility by 2 property since the transformation preserves parity.

Assume this is the first time we get (0,0,0). Then it must be a of the form $2a - x$. Where every component of x is divisible by 2 since every element of $2a$ is divisible by 2. Therefore every component of x is divisible by 2, since other wise the the product wouldn't have every component divisible by 2. But since none of the original starting points had every component divisible by 2, then must also a product of the form $2y - z$, where z 's every component is also divisible by 2. Using this logic we get an infinite chain of points, therefore you can't get (0,0,0) from the set of given points.

The book stated that the invariant is the parity of the component. It stated that you can't get (1,1,1), same mine but in reverse.

2.24 Problem 42

The following game is played on an infinite chessboard. Initially, each cell of an $n \times n$ square is occupied by a chip. A move consists in a jump of a chip over a chip in a horizontal or vertical direction onto a free cell directly behind it. The chip jumped over is removed. Find all values of n , for which the game ends with one chip left over (IMO 1993 and AUO 1992!).

Solution

I have been thinking about this and I am getting no where. The one thing I can think about are functions that are similar to the previous question about solitaire jumps.

But when I try to make a function that has an invariant everywhere, or atleastt weights assigned to the board, such that the total is conserved when jumping, I get the problem where if go back the weight decreases. Basically I want a function that models

the moves atleast well enough so an impossibility on the function implies and impossibility on the game.

I think it is impossible for all $n > 2$. I was thinking of having function that just increases outward proportionally exponentially to the manhattan distance from the origin. Leaving aside the issue of how I am going to define the origin, it feels like the functions I come up, don't work for this problem because a chip can move back and I don't have a function that will increase, in all directions, the closest thing I have is a function that attaches the manhattan distance weight. So basically the weight of a chip is $\phi^{-\|x\|-\|y\|}$, where ϕ is the positive root of $x^2 - x - 1 = 0$.

Now what do I do then? Should I go to polar coordinates? But I still experience the same problem of going back? Can I restrict so that a chip doesn't ever go back? Can I show that this doesn't effect the outcome of the game?

There is this sense that the chips need to be close together to be able to execute a move, but not so much that it restricts movement. Is there function that has this much control? Do I even need this much control?

What if I shift the origin, so that the function is decreasing in real time? Like everytime, the a move is made, the origin of the entire grid also changes, this way I may be able to combat the chip going back problem?

I feel that this is an interesting idea? which direction should the origin move? What happens if a chip moves? Actually it maybe the same problem as before? What if I assign weights to the chips instead of the positions?

But how would I incorporate the feasibility of moves into this definition of a function? Ah I need a hint or something.

Ok I may need to replay the game again.

For 1×1 , it is already done. What about 2×2 , I can move the bottom left, and eat the top left, and do the same for bottom right and I get two on the same row, so one eats another, and I get 1.

For 3×3 , how do I do this one? It actually turns out, I can't the closest I get is that the two chips are two units away? I smell some mod2 business? What really happens when a piece eats another? what happens to the position or the parity of the chips?

Let me try 4×4 . I get 3, chips next to each other. Is that the relation? For $n \times n$, square of chips, I get $n - 1$ chips, except the case of $n = 1$.

What invariants do I see then. Should I do the coordinates of the chips getting eaten. every time chip (x_1, y_1) eats Another chip next to it, then then mod 2, is the same? The before and after mod 2 of the eating chip, is the same?

Is this invariant over the whole board? What about the chip being eaten? What happens to last chip? It must be next to a chip of another parity? But one of the has to have eaten something and came there, so either the one with odd parity or even parity ate another chip of different parity. So by extension we either have odd and even, then odd or even. But this one must also have eaten another one. So a chip of another parity. The Must this be the case though? Could I have two moves of the same parity being eaten, like for example, one chip eats another chip to allow space for another chip of the same parity to eat and end up at that position?

Like me try to extend this idea of working backwards. The last two chips must have different parity and be next to each other. One of them must have eaten something and came here. So two chips of the same parity, and a different one.

What if I make an invariant on the parity of the total removed chips and the total chips on the board. Does this invariant tell me anything? Actually nothing much.

A move, moves a chip and removes a chip. If I consider the total parity of all chips both removed and on the table it doesn't change. If I started with $n \times n$, where even, then I must end in an even parity position. The last chip must have an even manhattan distance from the origin. But why though?

Ok after reading the book I realized that my approach was not going any where. The secret is that it only works for. n when n isn't divisible by 3. I should have tried more systematic ways of reduction. The proof in the book and online relied on a series of moves and techniques that reduce it the size by 3 in each dimension. Then you using a coloring proof for the case it is divisible by 3.

I need to get rid of this habit of looking of invariants. It will limit my creativity. I will list the proof for the impossibility of this problem when n is divisible by 3 for brevity. Actually the it is still mod2. Damn mod 2 rules everything. Ok, first assume n is divisible by 3.

Color diagonally all chips either A, B or C , and let a, b , and c , represent the number of chips on each color. Then everytime you move, two decrease will one increases, so $a \equiv b \equiv c \pmod{2}$. But this relation will be broken if you have only one color.

2.25 Problem 43

Nine 1×1 cells of a 10×10 square are infected. In one time unit, the cells with at least two infected neighbors (having a common side) become infected. Can the infection spread to the whole square?

Solution

I feel this has something to do with the perimeter? Oh, that is because is. I need to prove that the perimeter doesn't increase. When a cell gets infected, if it is touching two infected cells, it the two infected cells lose a perimeter and they gain two from the infected one, so atmost teh perimeter can't grow. Since at the start the perimeter is 36, it can't cover the full 40 unit perimeter.

2.26 Problem 44

Can you obtain the polynomial $h(x) = x$ from the polynomials $f(x)$ and $g(x)$ using only the operations of addition, subtraction, and multiplication in the following cases?

1. $f(x) = x^2 + x, \quad g(x) = x^2 + 2.$
2. $f(x) = 2x^2 + x, \quad g(x) = 2x.$
3. $f(x) = x^2 + x, \quad g(x) = x^2 - 2.$

Solution

So the question is asking can I get the Identity function my combining these two functions, through this methods?

I feel that every question is different and will probably require a different invariant, so I may just tablet to write. Ok reading the first explanation, I think I got the gist of it. So for the first one, if we put in $x = 2$, I always get a value that is a multiple of six, so I can't possible get 2.

For the second one. Through some combination, I need to get x , and this is valid for all x . What if I put in 2? If I combine them any, whether though every operation, they will still be divisible by 4, and that is't correct.

For the third one? What if I put in, an irrational number. I will not get an irrational back? like $\sqrt{2}$, unfortunately doesn't work.

After reading a book apparently there a mode that works. Let me see if I can derive it. I have to get rid of the 2nd order terms. Then I get $x + 2$. Ok after looking it is $(f - g)^2 + 2g - 3f = x \leftarrow x^2 + 4x + 4 + 2x^2 - 4 - 3x^2 - 3x = x$. How would I come up with that, so once I that I got that new quadratic multiply to get a new quadratic and then I guess do some linear algebra? Also I limited myself unnecessarily by only taking multiplication between functions.

2.27 Problem 45

Accumulation of Computer Rounding Errors

Start with $x_0 = 1$ and $y_0 = 0$. Using your computer, generate the sequences:

$$x_{n+1} = \frac{5x_n - 12y_n}{13}$$

$$y_{n+1} = \frac{12x_n + 5y_n}{13}$$

Compute and find $x_n^2 + y_n^2$ for:

$$n = 10^2, \quad 10^3, \quad 10^4, \quad 10^5, \quad 10^6, \quad 10^7.$$

Solution

OK the answer.

0.9999999999999992 0.9999999999999983 1.00000000000000027 0.9999999999999978 1.0000000000000000
0.999999999999996455

Why is that happening? What the hell is happening? I guess it overestimates and underestimates but I don't why this is happening in python I guess Python doesn't have arbitrary precision.

2.28 Problem 46

Start with two numbers 18 and 19 on the blackboard. In one step you may add another number equal to the sum of two preceding numbers. Can you reach the number 1994 (IIM)?

Startin with $x = 18, y = 19$

$(x, y) \rightarrow (y, x + y)$

Is this a deterministic process, there is nothing I can control?

So let me try, $(19, 37) \rightarrow (37, 56) \rightarrow (56, 93) \rightarrow (93, 149) \rightarrow (149, 242) \rightarrow (242, 391)$
 $(391, 633) \rightarrow (633, 1024) \rightarrow (1024, 1657) \rightarrow (1011, 1636) \rightarrow (1636, 2647)$

Ok what?

I miss understood the rules. I hate that the book isn't clear!!! The book had $1944 = 104 * 19 + 18$

2.29 Problem 47

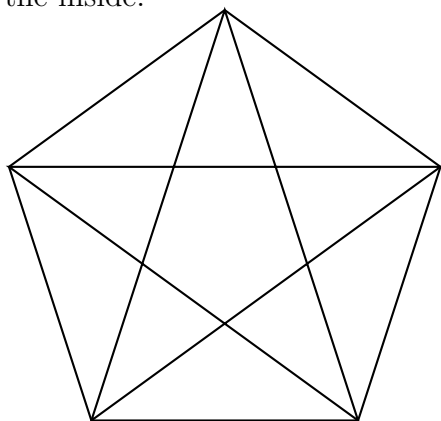
In a regular (a) pentagon (b) hexagon all diagonals are drawn. Initially each vertex and each point of intersection of the diagonals is labeled by the number 1. In one step it is

permitted to change the signs of all numbers of a side or diagonal. Is it possible to change the signs of all labels to -1 by a sequence of steps (IIM)?

Solution

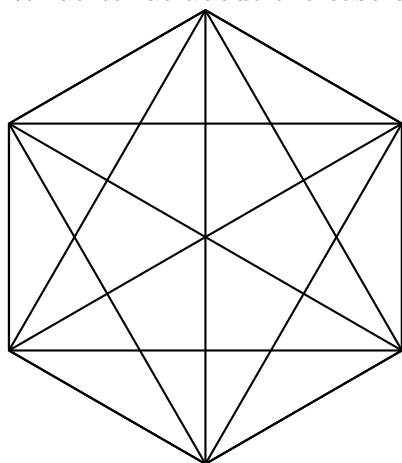
Ok, is this possible? Let me first try, then pentagon has 10 points, of one 1. And The diagonals have 3 vertices on them, and the edges have 2 vertices on them. Everytime I flip, an edge the number of 1 decreases or increases by 1,2 or 3 depending on which I choose. What about parity of the innner points and the digonals it is connected to? Can I change one of them without affecting the other.

Then vertex in the inside is connected to four vertices on the outside, Now I see what it was 5 on the inside. Ok so only one them the vertices isn't connected with the vertex on the inside.



I found the invariant, the sum of the vertices, actually just do an indicator function whether the vertex is 1 or 0. Everytime you dojjj a move, it always interacts with two of the vertices on the outside, and preserves the parity, the case of having all -1, corresponds to having 0, but that has different parity from 5, so it is impossible.

What What about the case of a hexagon?



I must say it looks, beautiful. It feels like there is so much freedom. But there are 19 points in total. Now I am feeling extra fishy. Let me think on it.

Ok I couldn't find a solution using the same trick as before? What about a different? The function I make must have some nonzero coefficient for the center. There is also the problem that there moves that intersect 2, and 5 nodes. And I would have to treat the center the same as the ones on the middle of the side of the center hexagons.

Maybe it is possible....

After reading the book, it said the product of the number on the nine circles doesn't change?

What did I learn? Consider product when doing invariants, especially mod 2, and any collection of odd objects with even intersection works.

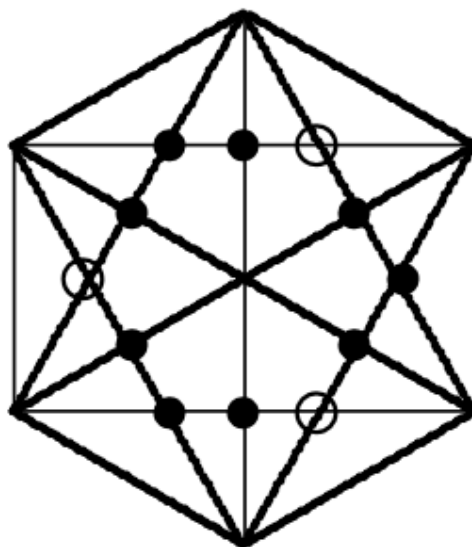


Fig. 1.11

2.30 Problem 48

In Fig. 1.7, two squares are neighbors if they have a common boundary. Consider the following operation T : Choose any two neighboring numbers and add the same integer to them. Can you transform Fig. 1.7 into Fig. 1.8 by iteration of T ?

Fig. 1.7

$$\begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}$$

Fig. 1.8

$$\begin{bmatrix} 7 & 8 & 9 \\ 6 & 2 & 4 \\ 3 & 5 & 1 \end{bmatrix}$$

Solution

This is the checkerboard problem, first color it like a checker board and see if the difference in the total sum of all white and black cells of the first box matches that of the

second. Then if they do match you turn the first one by first fixing the top left 2×2 box and you can do this first fixing the top left and then using the same technique to go down. Then in the last row fix every number and the last two must be the same in the box since the difference has to be the same so you can either increase the two numbers to match or it decrease them.

I don't want to check specifically, oh well I will then.

The first one is (top left is assumed to be black): $1+3+5+7+9$, the white is $2+4+6+8=15$, $20-15=5$. The second one is: $7+9+2+3+1$, white is $8+4+6+5$. So the difference is $22-23=-1$.

Unfortunately $-1 \neq 5$, basically the invariant is the difference in the sum of white squares and black squares.

2.31 Problem 49

There are several signs $+$ and $-$ on a blackboard. You may erase two signs and write, instead, $+$ if they are equal and $-$ if they are unequal. Then, the last sign on the board does not depend on the order of erasure.

Solution

Is the product just conserved? It literally is!

Ok could I do it with XOR? Isn't that also just checking the parity as well. I guess there is nothing then.

2.32 Problem 50

There are several letters e , a and b on a blackboard. We may replace two e 's by one e , two a 's by one b , two b 's by one a , an a and a b by one e , an a and an e by one a , a b and an e by one b . Prove that the last letter does not depend on the order of erasure.

Solution

I have to make the recurrence function. I also need to write down the functions because this is just too confusing.

a is the number of a 's, and henceforth for all the other letters.

$$\begin{aligned}(a, b, e) &\rightarrow (a, b, e - 1) \\(a, b, e) &\rightarrow (a - 2, b + 1, e) \\(a, b, e) &\rightarrow (a + 1, b - 2, e) \\(a, b, e) &\rightarrow (a - 1, b - 1, e + 1)\end{aligned}$$

The question has a lot of detours since the three of the operations are the same.

The difference between a and b , mod 3, doesn't change so the if the difference between them mod 3, then the final choice has to be e . Or basically $S = a - b \pmod 3$ is invariant. But I need to find the values of the final chip when it isn't the case that the difference isn't mod 3 is 0. What if I do $S = 2a + 2b \pmod 3$, when a decreases by 2, and b increases by 1, does it still become invariant? a 's change becomes a decrease of 1 mod 3, and b 's increase becomes 2 in mod 3. So S should stay invariant. Actually this doesn't matter since I can't use this to tell apart a and b . Since decreasing from a and decreasing from b are symmetric, their iterations must be similar, and their coefficients equal. Actually I

think $S = a - b$ suffices. If $S = 0$, then it is e, if 1 then it is a, if 2, then it is b. This is also a criteria for there to only exist one chip left in the end.

How do I make sure this is true. Since the invariant is constant, if there is a one chip at the end, if it is e, then S must have been 0. If S=1, then there is one a chip and if it is 2, then that means there is one b chip since $2 \equiv -1 \pmod{3}$.

Actually after reading the book it had a very cool proof.

$$e \circ e = e, \quad e \circ a = a, \quad e \circ b = b, \quad a \circ a = b, \quad b \circ b = a, \quad a \circ b = e.$$

The $\circ$ operation is commutative since we did not mention the order. It is easy to check that it is also associative, i.e.,

$$(p \circ q) \circ r = p \circ (q \circ r)$$

for all letters occurring. Thus, the product of all letters is independent of the order in which they are multiplied.

2.33 Problem 51

A dragon has 100 heads. A knight can cut off 15, 17, 20, or 5 heads, respectively, with one blow of his sword. In each of these cases, 24, 2, 14, or 17 new heads grow on its shoulders. If all heads are blown off, the dragon dies. Can the dragon ever die?

Soltion

Let h represent the number of heads.

So the operations I can do are

$$\begin{aligned} h \wedge h \geq 15 &\rightarrow h + 9 \\ h \wedge h \geq 17 &\rightarrow h - 15 \\ h \wedge h \geq 20 &\rightarrow h - 6 \\ h \wedge h \geq 17 &\rightarrow h + 15 \end{aligned}$$

I think is also dies if the number of heads is less or equal to 20. Since I can hit it and it will die.

If the requirement is only all of the heads are blown off until it dies, then it can be killed by just used blowing 20 heads each time.

If exactly 0 heads must exist at a time then it is impossible because it has each move preserves the mod 3 parity, and the number of heads I must chop off need to have 0, and 2 and mod 3 but 100 has 1 mod 3.

2.34 Problem 52

Is it possible to arrange the integers $1, 1, 2, 2, \dots, 1998, 1998$ such that there are exactly $i - 1$ other numbers between any two i 's?

Solution

This is like the codeforce blog. I will think about it later because is 3:11 PM, and I need to sleep.

So let's say the indices of the numbers are $i_{11}, i_{12}, i_{21}, i_{22}, \dots$. The difference added up should give the below expression.

$$(i_{21} - i_{11}) + (i_{22} - i_{21}) + \cdots = (0) + (1) + (2) + \cdots + 1997$$

If you take this mod 2, then $a - b \equiv a + b \pmod{2}$. So

$$i_{11} + \cdots = \frac{1998 * 1997}{2} = 999 * 1997$$

Since the sum of all the doubles of the numbers on the left is odd, the sum is even and the one on the right is wrong therefore the sequence can't exist.

Would this proof for cases where I was asked an odd number? Actually it depends on the mod 4. When the final value mod 4 is 3, then once I add one and divide I get an odd number, therefore the proof works in that case. Wait I mean when the final number(1998) mod 4 is 2, I can do this proof. So for all numbers with mod 4, you can't do this process, what about 0,1,3? In the case of 0, it will be divisible and the proof will not work. In the case of 1? It will also fail, in the case of 3? It will work. SO for all numbers who have mod 4, 2,3 and this will proof will work, and it is impossible.

2.35 Problem 53

The following operations are permitted with the quadratic polynomial $ax^2 + bx + c$:

- (a) Switch a and c .
- (b) Replace x by $x + t$, where t is any real number.

By repeating these operations, can you transform $x^2 - x - 2$ into $x^2 - x - 1$?

Solution

The discriminant is conserved, I did a question like this before. The discriminant of the two quadratics aren't the same. The book actually has an elegant point about what the discriminant is. It is the difference between, and another insight I didn't think about before is that the min or max of the quadratic function is between the roots, so $\frac{-b}{2a} = \frac{x_1+x_2}{2} \rightarrow \frac{-b}{a} = (x_1 + x_2)$.

This is an invariance problem. As a prime candidate, we think of the discriminant D . The first operation obviously does not change D . The second operation does not change the difference of the roots of the polynomial.

Now,

$$D = b^2 - 4ac = a^2 \left(\left(\frac{b}{a} \right)^2 - 4 \frac{c}{a} \right),$$

but

$$-\frac{b}{a} = x_1 + x_2, \quad \text{and} \quad \frac{c}{a} = x_1 x_2.$$

Hence,

$$D = a^2(x_1 - x_2)^2,$$

i.e., the second operation does not change D .

Since the two trinomials have discriminants 9 and 5, the goal cannot be reached.

2.36 Problem 54

Initially, we have three piles with a , b , and c chips, respectively. In one step, you may transfer one chip from any pile with x chips onto any other pile with y chips.

Let

$$d = y - x + 1.$$

If $d > 0$, the bank pays you d dollars. If $d < 0$, you pay the bank $|d|$ dollars.

Repeating this step several times, you observe that the original distribution of chips has been restored.

What is the maximum amount you can have gained at this stage?

Solution

So d is the old value of x - new value of y . Does it have any deeper meaning? Ok, what should I do then? I want find the maximum amount of money I got? I feel that this should be 0, but I am not sure let me think about, I will sleep now, since it is 3:19.

I couldn't figure the how it actually worked but, the invariant is $a^2 + b^2 + c^2 - 2 * g$. How should I be able to recognize? the combination should have looked, and I should look for invariants to make it zero.

2.37 Problem 55

Let $d(n)$ be the digital sum of $n \in \mathbb{N}$. Solve the equation:

$$n + d(n) + d(d(n)) = 1997.$$

Solution

Since the digit summed is conserved under module 9, and the we have to be able to find the inverse of $1997 \bmod 9$, which is the inverse of 8 in mod 9. Actually we want something times 3, to give 8 in module 9. So it is impossible, the book said the transformation is invariant under modul 3. Actually I get it now, since the sum of the digits is used to check divisibility for 3, which we can use it to know that d is invariant under module 3. So the equation becomes $0 \equiv 2 \bmod 3$, which is impossible. That was simpler than my proof.

2.38 Problem 56

Start with four congruent right triangles. In one step, you may take any triangle and cut it in two with the altitude from the right angle. Prove that you can never get rid of congruent triangles.

Solution

I actually couldn't solve this question, it was quite tricky to get the formulas correct, the book's insight was that you can describe the first triangle as $1, p, q$ where $1 \geq q, p$ and then all the next triangles will be with coefficients of $p^m q^n$. Then you can use the same lattice point with powers of 2 technique in question 39, to show that you can't have four all the points be distinct on the lattice points without filling up the entire plane with would require infinite moves.

2.39 Problem 57

Starting with a point $S(a, b)$ of the plane with $0 < a < b$, we generate a sequence (x_n, y_n) of points according to the rule:

$$x_0 = a, \quad y_0 = b$$

$$x_{n+1} = \sqrt{x_n y_{n+1}}, \quad y_{n+1} = \sqrt{x_n y_n}.$$

Prove that there is a limiting point with $x = y$. Find this limit.

Solution

What could the invariant be? Could it be in the example with the product being fixed and the end result having something to do with cosine?

The value $x_{n+1}\sqrt{y_{n+1}} = x_n\sqrt{y_n}$. The values of y are decreasing and the values of x are increasing, and the values stay constant after reaching $x = y = (a^2b)^{1/3}$.

Actually I didn't rigorously prove that x is increasing and y is decreasing. But the book had an interesting point on the relationship between the geometric mean, and the arithmetic mean. The arithmetic mean is always bigger.

2.40 Problem 58

Consider any binary word $W = a_1a_2 \cdots a_n$. It can be transformed by inserting, deleting, or appending any word XXX , where X is any binary word. Our goal is to transform W from 01 to 10 by a sequence of such transformations.

Can the goal be attained? (LMO 1988, oral round)

Solution

Can I do this? The ones I remove will have to be a 3 repetition and the ones I add will also have to be 3 repetitions. What is conserved when I do this operation?